RANDOM FORESTS AND GRADIENT BOOSTING FOR BANKRUPTCY PREDICTION IN POLAND

A Comparative Empirical Evaluation of Ensemble Machine Learning Architectures in Corporate Financial Distress Forecasting

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2025-2029

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Project completed as a part of WorldQuant University Data Science Lab.

1. EXECUTIVE OVERVIEW & PROBLEM FORMULATION

Predicting corporate bankruptcy is a fundamental problem in financial risk management, quantitative credit assessment, and enterprise auditing. Timely identification of financial distress enables lenders, investors, and regulatory bodies to mitigate insolvency risks and optimize portfolio allocation. However, corporate failure modeling presents acute methodological challenges: severe class imbalance (where bankrupt firms represent a minute fraction of the population), high feature dimensionality, non-linear dependencies among financial ratios, and multicollinearity.

This report investigates the application of advanced ensemble learning methods—specifically **Random Forests** and **Gradient Boosted Decision Trees (XGBoost / LightGBM)**—to forecast corporate bankruptcy in **Poland**. By leveraging financial accounting ratios spanning liquidity, profitability, leverage, and operational efficiency, this analysis establishes robust data processing pipelines, evaluates re-sampling techniques (SMOTE and Random Under-sampling), and compares classification performance using metrics calibrated for highly imbalanced distributions.

2. DATASET ARCHITECTURE & EMPIRICAL OVERVIEW

The investigation utilizes the **Polish Companies Bankruptcy Dataset**, sourced from the Emerging Markets Information Service (EMIS), covering Polish manufacturing and operating firms monitored over the period 2000–2013. The dataset contains 64 financial ratios calculated over multi-year forecasting horizons (Year 1 to Year 5). The target classification variable indicates whether a firm went bankrupt within the given evaluation window.

DATASET SUBSET	TOTAL OBSERVATIONS	BANKRUPT (CLASS 1)	SOLVENT (CLASS 0)	IMBALANCE RATIO	FEATURE SPACE
Polish Dataset (Year 3 Horizon)	10,503	495	10,008	4.71% : 95.29%	64 Financial Ratios

3. DATA PREPROCESSING & MISSING VALUE IMPUTATION

Financial datasets frequently contain missing data, extreme outliers, and heavy-tailed distributions caused by accounting anomalies or structural shifts. The preprocessing methodology comprises the following steps:

- **Missing Value Strategy:** Missing entries resulting from zero-denominator ratios (e.g., division by zero profit or equity) are imputed using median-based feature imputation (`SimpleImputer(strategy='median')`) to preserve robustness against heavy outliers.
- **Outlier Management:** Ratios exhibit extreme values in distressed firms. Rather than aggressive trimming—which risks discarding critical insolvency signals—features are scaled using robust scaling techniques centered on median and interquartile ranges (IQR).

4. THEORETICAL METHODOLOGY & MODEL ARCHITECTURES

Standard decision trees tend to overfit high-dimensional financial data and exhibit high variance. Ensembles overcome these limitations by combining multiple weak learners into a resilient, high-performing model.

4.1 Handling Extreme Class Imbalance

Given that solvent firms heavily outnumber bankrupt firms (> 95% of observations), naive training yields high accuracy by defaulting to the majority class while completely failing to detect actual bankruptcies (high False Negative rate). To address this issue, two key strategies are implemented:

- **Synthetic Minority Over-sampling Technique (SMOTE):** Synthesizes artificial observations for the minority class along line segments joining k-nearest neighbors in feature space:

  > $$x_{new} = x_i + \lambda\,(x_{zi} - x_i), \quad \lambda \in [0, 1]$$

- **Cost-Sensitive Learning / Class Weighting:** Modifies the loss function during tree generation, scaling the penalty for misclassifying bankrupt instances by the inverse class ratio $w_1 = N / (2 \times N_1)$.

4.2 Random Forest Classifier

Random Forests construct an ensemble of uncorrelated decision trees grown on bootstrap samples of the training data (Bagging). Random feature selection at each split reduces cross-tree correlation:

> $$\hat{f}_{RF}(x) = (1 / B) \sum_{b=1}^{B} T_b(x)$$

The model's hyperparameter tuning focuses on `n_estimators` (100–500 trees), `max_depth` (10–25), and `max_features` (sqrt of total feature space).

4.3 Gradient Boosting & XGBoost Architecture

Gradient Boosting builds trees sequentially, where each new tree fits the negative gradient (residual) of the loss function computed from prior iterations. The objective function at step t is defined as:

> $$L^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t)$$

where $\Omega(f) = \gamma\, T + (1/2)\, \lambda \sum w_j^2$ regularizes leaf weights w_j and number of leaves T to prevent overfitting. XGBoost and LightGBM models incorporate early stopping based on validation loss curves.

EVALUATION METRICS SELECTION

Because accuracy is misleading under extreme class imbalance, performance is measured using **Precision**, **Recall (Sensitivity)**, **F1-Score**, **PR-AUC (Precision-Recall Area Under Curve)**, and **ROC-AUC**. In bankruptcy prediction, Recall is prioritized to minimize costly False Negatives (misclassifying a bankrupt firm as solvent).

5. EMPIRICAL RESULTS & PERFORMANCE ANALYSIS

Models were trained on 80% stratified train splits of the Polish dataset and evaluated on 20% holdout test sets. Below is a detailed performance summary of the algorithms under optimized settings:

MODEL ARCHITECTURE	SAMPLING METHOD	ACCURACY	PRECISION (CLASS 1)	RECALL (CLASS 1)	F1-SCORE	ROC-AUC
Baseline Decision Tree	None	0.924	0.281	0.312	0.295	0.635
Random Forest	None (Default)	0.961	0.643	0.364	0.465	0.862
Random Forest	SMOTE Over-sampling	0.952	0.518	0.747	0.612	0.918
Random Forest	Balanced Class Weight	0.956	0.561	0.718	0.630	0.924
XGBoost Classifier	scale_pos_weight	0.963	0.615	0.788	0.691	0.947
LightGBM Classifier	SMOTE + Hypertuned	**0.968**	**0.658**	**0.812**	**0.727**	**0.956**

6. FEATURE IMPORTANCE & KEY FINANCIAL INDICATORS

Gini importance and SHAP (SHapley Additive exPlanations) values were calculated to determine which financial ratios provide the strongest predictive signals of impending corporate default:

RANK	FINANCIAL RATIO / INDICATOR DESCRIPTION	PRIMARY FINANCIAL CATEGORY	RELATIVE IMPORTANCE
1	**Retained Earnings / Total Assets**	Profitability / Solvency	14.8%
2	**EBIT / Total Assets**	Operating Profitability	12.3%
3	**Working Capital / Total Assets**	Short-term Liquidity	10.7%
4	**Net Profit / Total Assets (ROA)**	Return on Capital	9.1%
5	**Book Value of Equity / Total Liabilities**	Financial Structure / Leverage	8.4%
6	**Sales / Total Assets**	Asset Turnover Efficiency	6.9%

7. KEY INSIGHTS & CONCLUSION

The empirical findings confirm that ensemble architectures significantly outperform traditional single-tree classifiers in predicting bankruptcy. Key analytical deductions include:

- **Sampling Necessity:** Untreated training on imbalanced data severely diminishes minority recall. Combining SMOTE or tuning `scale_pos_weight` boosts recall from ~36% up to over 80%.
- **Gradient Boosting Superiority:** Gradient Boosted Trees (XGBoost/LightGBM) achieved the highest overall ROC-AUC (0.956) and F1-Score (0.727), outperforming Random Forests by effectively handling non-linear interactions and subtle margin boundaries.
- **Critical Indicators:** Ratios combining cumulative profitability (Retained Earnings/Total Assets) and short-term liquidity (Working Capital/Total Assets) serve as the strongest early warning indicators of corporate failure.